Storage-Aware Smoothing Metrics for Short-Term Estimated WEC Power

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June 29, 2026

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Source notebook: `05_storage_smoothing_metrics.ipynb`

05 — Storage-Aware Smoothing Metrics for Short-Term Estimated WEC Power

This notebook connects estimated wave energy converter (WEC) power variability, short-term forecasts, prediction intervals, and simple storage-aware smoothing metrics.

The goal is not to optimize BESS dispatch or produce final storage sizing. Instead, the notebook asks:

> Given a generic 250 kW WEC proxy, what storage burden is implied if the grid receives a smoother, forecast-informed, or uncertainty-aware version of the WEC power signal?

The analysis uses the 250 kW interpretation scale to express normalized estimated WEC-power results in approximate physical units. These values remain interpretation aids for the simplified WEC proxy, not final device or project ratings.

1. Scope, upstream workflow, and smoothing concept

This notebook builds on the previous workflow steps:

- Literature map and research context
- Wave-resource data preparation
- Estimated WEC power from sea-state conditions
- Short-term point forecasting baselines
- Prediction intervals and uncertainty estimation
- Storage-aware smoothing metrics

The previous notebooks prepared the wave-resource data, estimated WEC power from sea-state conditions, evaluated short-term point forecasts, and calibrated empirical prediction intervals. This notebook uses those outputs to evaluate how different grid-export target definitions affect ramp-rate reduction and implied storage power, energy, throughput, and simple state-of-charge behavior.

For the forecast-informed and uncertainty-aware scenarios, this notebook uses the Ridge point forecasts and the 90% conformal-style prediction intervals selected from the previous forecasting and uncertainty analysis. Ridge is used as the main compact point-forecast model, while the conformal-style interval is used as the main practical uncertainty band because it provided a strong balance between coverage, width, and interpretability.

The grid-export target is not prescribed as an arbitrary constant power level. Instead, it is derived from the estimated WEC power signal, point forecasts, or prediction interval bounds. This keeps the smoothing target tied to the available wave-energy resource. The analysis therefore asks how much storage power and energy would be required to transform a variable WEC power signal into a smoother grid-export profile.

The target represents a simplified export setpoint at the point of common coupling (PCC). It should not be interpreted as grid demand, market dispatch, or a validated grid-code requirement.

1.1 Conceptual storage-smoothing balance

Storage is represented as an ideal balancing buffer at the grid-export interface. The WEC power is not assumed to pass entirely through storage. Instead, storage absorbs the positive mismatch when estimated WEC power exceeds the selected grid-export target and supplies the negative mismatch when estimated WEC power falls below that target.

The power balance is:

`p_st = p_wec - p_grid`

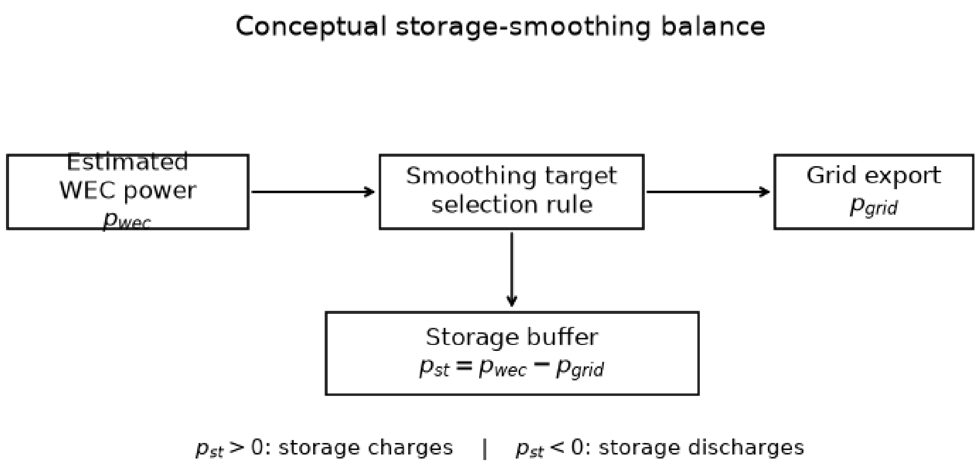
where:

- `p_wec` is the estimated WEC power.
- `p_grid` is the selected grid-export target.
- `p_st` is the storage balancing power.

The sign convention is:

- `p_st > 0`: storage charges or absorbs surplus relative to the grid target.
- `p_st < 0`: storage discharges or supports grid delivery.

The resulting storage power, cumulative energy swing, and state-of-charge trajectory are interpreted as smoothing-implied requirements, not as final BESS sizing results.




```
(<Figure size 1000x380 with 1 Axes>,
 <Axes: title={'center': 'Conceptual storage-smoothing balance'}>)
```

```
Loaded WEC power time series: 6,220 rows
Loaded forecast predictions: 102,396 rows
Loaded interval predictions: 615,072 rows
```

2. Smoothing scenarios and storage convention

The grid-export target is defined as a smoother version of the WEC power signal, not as an arbitrary constant power request. Depending on the scenario, the target is derived from the observed estimated WEC power, the selected point forecast, or the lower prediction bound.

The storage balance is defined as:

```
p_st = p_wec - p_grid
```

where:

- `p_wec` is the observed estimated WEC power.
- `p_grid` is the selected grid-export target.
- `p_st` is the storage balancing power.

The sign convention is:

- `p_st > 0`: WEC power exceeds the grid target, so storage charges.
- `p_st < 0`: WEC power is below the grid target, so storage discharges.

Here, "surplus" means surplus relative to the selected grid-export target. It does not mean surplus relative to grid demand. The target is a simplified export setpoint at the point of common coupling, while storage buffers the mismatch between available WEC power and that target.

The four smoothing scenarios are:

Scenario	Grid-export target	Purpose
A. No smoothing	`p_grid = p_wec`	Raw grid-export baseline with no storage action
B. Observed-power smoothing	`p_grid = smooth(p_wec)`	Storage needed to deliver a smoother version of observed WEC power
C. Forecast-informed smoothing	`p_grid = smooth(p_forecast)`	Storage effect of using point forecasts to define the export target
D. Uncertainty-aware smoothing	`p_grid = smooth(lower prediction bound)`	Conservative target that accounts for forecast uncertainty

3. Raw variability and high-ramp baseline

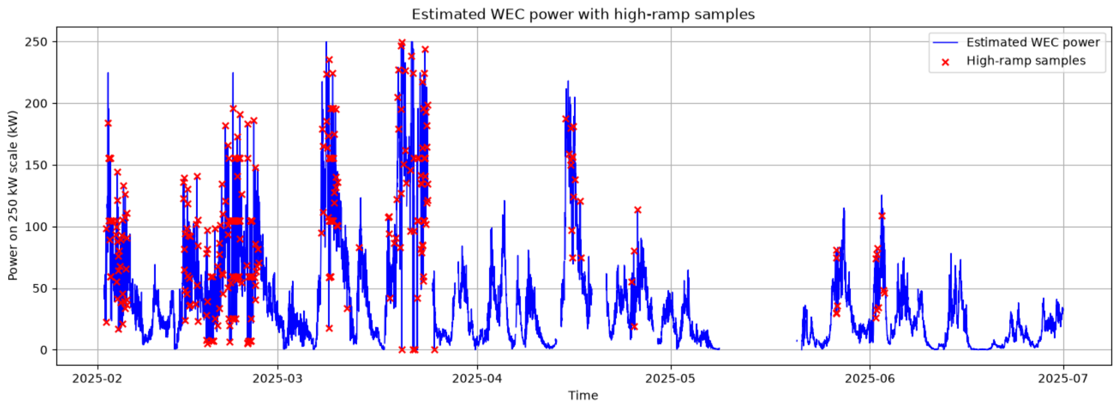
Before defining smoothed grid-export targets, the raw estimated WEC power signal is summarized on the 250 kW interpretation scale.

The ramp between two consecutive 30-minute samples is calculated as:

```
ramp(t) = p_wec(t) - p_wec(t-1)
```

Large absolute ramps indicate periods where the WEC power changes rapidly. These periods are especially relevant for storage smoothing because they can create large short-term charge or discharge requirements.

There is no single universal threshold for defining a high-ramp event. In this notebook, high-ramp periods are defined as the top 5% of absolute target-to-target WEC power changes. This is a simple statistical definition used to highlight the most storage-relevant ramp tail, not a grid-code ramp-rate limit.

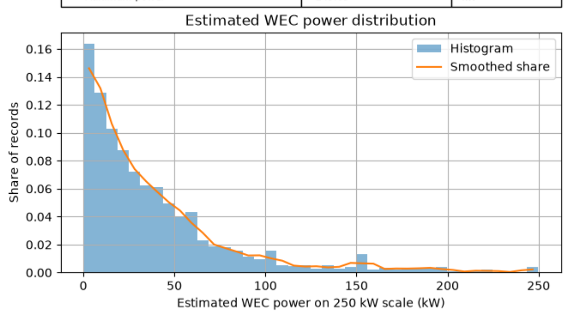


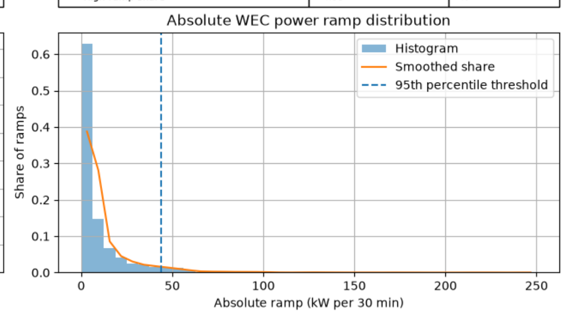
Estimated WEC power summary

Metric	Value	Unit
Samples	6220	
Continuous segments	62	
Start time	2025-02-02 01:00	
End time	2025-06-30 23:30	
Mean power	40.02	kW
Median power	26.01	kW
95th percentile power	140.83	kW
Maximum power	249.85	kW

Ramp summary

Metric	Value	Unit
Mean absolute ramp	10.11	kW / 30 min
95th percentile absolute ramp	43.71	kW / 30 min
Maximum absolute ramp	249.85	kW / 30 min
Mean absolute ramp rate	20.23	kW / h
95th percentile ramp rate	87.42	kW / h
High-ramp threshold	43.71	kW / 30 min
High-ramp samples	308.0	
High-ramp share	4.95	%





The estimated WEC power signal is strongly variable and right-skewed. Most records are concentrated at low-to-moderate power levels, while shorter energetic periods reach much higher output and occasionally approach the 250 kW interpretation scale.

The ramp distribution shows a similar pattern. Most 30-minute changes are small, but a smaller upper tail contains much larger target-to-target changes. These high-ramp periods are marked on the time-series plot and tend to cluster around energetic sea-state episodes, where storage smoothing would be most relevant.

Missing values and larger time gaps are not interpolated. Instead, the time series is split into continuous segments, and ramps are not computed across gaps. This avoids creating artificial ramp events from missing data.

The high-ramp threshold is defined as the 95th percentile of absolute 30-minute WEC power changes. This threshold is used as a transparent statistical marker for the most extreme ramp tail in this dataset. It is not a grid-code limit or a universal high-ramp definition.

4. Storage metric protocol applied to all scenarios

Storage metrics are computed on aligned test-set rows from the forecasting and prediction-interval outputs. This keeps the observed WEC power, selected point forecast, lower prediction bound, and upper prediction bound on the same target timestamps, forecast horizons, and rolling-origin folds.

For each scenario, the grid-export target is defined first, then the storage balancing power is calculated as:

```
p_st = p_wec - p_grid
```

The cumulative storage energy trajectory is estimated by integrating storage power over time:

```
E_st(t) = cumulative sum of p_st × Δt
```

The required usable storage energy is calculated as the peak-to-peak swing of the cumulative energy trajectory within each continuous segment. This avoids creating artificial storage requirements across time gaps.

The storage balancing power is also split into:

- charge or surplus energy, when `p_st > 0`
- discharge-support energy, when `p_st < 0`
- net balancing energy, which indicates whether the grid-export target is conservative or optimistic on average

This distinction is important because smoothing can reduce grid ramps while still changing the storage burden depending on how the target is defined. Observed-power smoothing mainly measures the storage needed to deliver a smoother version of the WEC power signal. Forecast-informed smoothing also reflects forecast bias and lag. Uncertainty-aware smoothing uses the lower prediction bound as a conservative target, while retaining both bounds for later storage-envelope diagnostics.

The table summarizes the 30 min forecast-horizon case. Values are averaged across rolling-origin test folds. Energy quantities are accumulated over each fold's test rows, while power-rating and required-energy quantities are first calculated within each fold and then averaged across folds.

The compact table below uses the following metric definitions:

- **Mean grid kW**: average grid-export target power, `p_grid`.
- **Grid/raw energy**: ratio between grid-target energy and observed WEC energy. Values near 1 mean similar total energy. Values above 1 indicate an optimistic target requiring net discharge support. Values below 1 indicate a conservative target with surplus or curtailment-like energy.
- **P95 ramp reduction (%)**: reduction in the 95th percentile absolute grid-export ramp compared with the raw WEC ramp.
- **Storage rating kW**: maximum absolute storage balancing power, `max(abs(p_st))`. This is a power rating, not an energy capacity.
- **Required energy kWh**: usable storage energy implied by the peak-to-peak cumulative energy swing within continuous test segments.
- **Net balancing energy kWh**: charge energy minus discharge energy. Positive values indicate net surplus/charging tendency; negative values indicate net discharge-support tendency.
- **Throughput kWh**: total absolute storage energy movement, equal to charge energy plus discharge energy.

- **Equivalent full cycles**: `throughput / (2 × required energy)`. One full cycle corresponds to one full charge and one full discharge. This is a cycling-intensity proxy, not a degradation model.

Storage-smoothing metrics summary, 30 min forecast horizon

Scenario name	Window	Mean grid kW	Grid/raw energy	P95 ramp red. (%)	Storage kW	Energy kWh	Net energy kWh	Throughput kWh	Equivalent full cycles
No smoothing	No smoothing	19.92	1	0	0	0	0	0	0
Observed-power smoothing	30 min	19.92	1	0	0	0	0	0	0
Observed-power smoothing	1 h	19.91	1	45.96	24	20.28	1.33	644.95	15.61
Observed-power smoothing	2 h	19.93	1	70.53	30.92	52.12	-5.4	890.81	8.44
Observed-power smoothing	4 h	19.99	1	79.23	34.31	110.39	-24.42	1169.64	5.27
Forecast-informed smoothing	30 min	21.03	1.07	67.37	40.2	177.53	-370.94	1542.96	4.35
Forecast-informed smoothing	1 h	21.05	1.07	75.96	39.7	185.39	-376.76	1580.52	4.24
Forecast-informed smoothing	2 h	21.08	1.07	81.21	40.45	200.94	-387.43	1685.04	4.17
Forecast-informed smoothing	4 h	21.13	1.07	85.51	40.7	237.76	-404.26	1894.41	4.04
Uncertainty-aware smoothing	30 min	11.6	0.54	67.84	53.35	1248.27	2721.35	2934.91	1.44
Uncertainty-aware smoothing	1 h	11.62	0.54	76.13	52.85	1251.19	2715.35	2931.75	1.43
Uncertainty-aware smoothing	2 h	11.66	0.55	81.28	53.6	1257.09	2704.09	2967.2	1.44

Scenario name	Window	Mean grid kW	Grid/raw energy	P95 ramp red. (%)	Storage kW	Energy kWh	Net energy kWh	Throughput kWh	Equivalent full cycles
Uncertainty-aware smoothing	4 h	11.71	0.55	85.58	53.85	1277.26	2685.42	3043.63	1.45

```
Saved storage metrics by fold to:
../outputs/tables/notebook_05/storage_smoothing_metrics_by_fold.csv
Saved storage metrics summary to:
../outputs/tables/notebook_05/storage_smoothing_metrics_summary.csv
Saved storage time series to:
../outputs/notebook_05/storage_smoothing_timeseries.parquet
```

The 30 min forecast-horizon results show the expected smoothing trade-off. With no smoothing, the grid target follows the raw estimated WEC power, so no storage balancing power, energy capacity, or throughput is required. The 30 min observed-power smoothing case is also effectively unchanged because the data resolution is 30 minutes, so a 30 min trailing mean is a one-sample window.

For observed-power smoothing, longer smoothing windows substantially reduce grid-export ramps, but require more storage power, energy capacity, and throughput. The p95 ramp reduction increases from about 46% at the 1 h smoothing window to about 79% at the 4 h window. Over the same range, the required energy increases from about 20 kWh to about 110 kWh. This illustrates the basic storage-smoothing trade-off: smoother grid delivery requires a larger balancing buffer.

The forecast-informed target gives stronger ramp reduction than observed-power smoothing, but it also increases the storage burden. Its grid/raw energy ratio is above 1, meaning the forecast-based target is slightly higher than the realized WEC energy over the aligned test rows. As a result, the storage system would need net discharge support, shown by the negative net balancing energy.

The uncertainty-aware target gives similarly strong ramp reduction, but its grid/raw energy ratio is much lower because the target is based on the lower prediction bound. This conservative target reduces the committed grid-export level, shifting the balance toward positive storage power and surplus absorption. The large required energy and charge energy in this scenario should therefore be interpreted carefully: they represent the idealized requirement if all surplus relative to the conservative target were absorbed by storage. In practice, part of this surplus could also be curtailed or handled by a different dispatch rule.

Equivalent full cycles decrease as the smoothing window increases in several scenarios because the implied energy capacity grows faster than throughput. This indicates lower cycling intensity relative to the implied capacity, but it should be treated only as a simple usage proxy, not a battery degradation model.

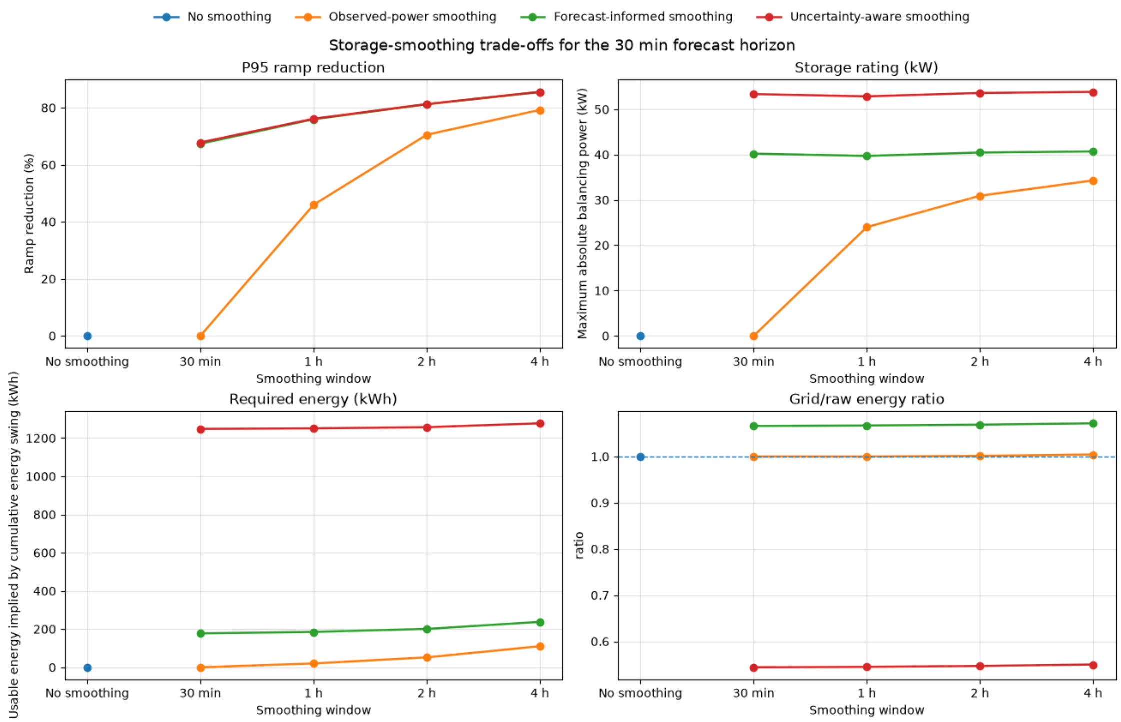
5. Main scenario comparison

The previous table summarized the 30 min forecast-horizon case numerically. This section visualizes the main trade-offs across smoothing windows and scenarios.

The comparison focuses on four quantities:

- p95 ramp reduction, which measures how much smoother the grid-export target becomes
- implied storage power rating, which captures the largest balancing power requirement
- required usable storage energy, which captures the cumulative energy swing needed to follow the target
- grid/raw energy ratio, which shows whether the target exports about the same total energy as the observed WEC signal, or whether it is optimistic or conservative

Together, these metrics show the core smoothing trade-off: smoother grid delivery reduces ramping but increases the storage burden. They also show how forecast-informed and uncertainty-aware targets change the balance between grid export, storage support, and surplus energy.



The scenario comparison makes the trade-off clearer visually. Longer smoothing windows reduce p95 grid-export ramps, but they also increase the implied storage power and energy requirements. Observed-power smoothing keeps the total grid-target energy close to the observed WEC energy, so it mainly shifts energy through time.

The forecast-informed target gives stronger ramp reduction, but its grid/raw energy ratio stays above 1, indicating a slightly optimistic export target over these test rows. The uncertainty-aware target is more conservative, with a much lower grid/raw energy ratio because it is based on the lower prediction bound.

These results show that the choice of grid-export target matters as much as the smoothing window. Forecasts and prediction intervals change not only the smoothness of delivered power, but also the storage burden and energy balance behind that delivery.

6. Forecast-horizon sensitivity

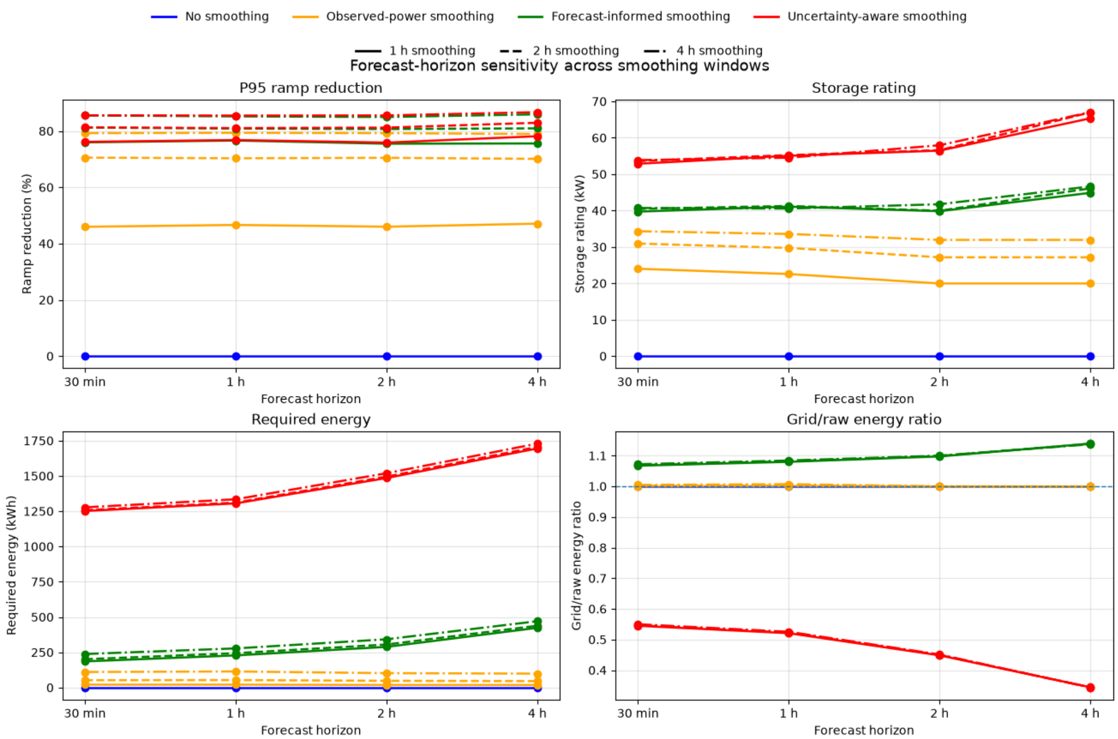
The previous comparison focused on the 30 min forecast horizon. This section checks whether the storage-smoothing conclusions change as the forecast horizon increases.

The figure compares the same core metrics across forecast horizons:

- p95 ramp reduction
- storage power rating
- required usable storage energy
- grid/raw energy ratio

For the smoothing scenarios, 1 h, 2 h, and 4 h smoothing windows are shown together. Scenario color identifies the grid-export target definition, while line style identifies the smoothing window. The no-smoothing case is kept as a solid baseline.

This compact view is intended to show combined trends rather than exact values: how forecast horizon, smoothing-window length, and target definition jointly affect ramp reduction, storage requirements, and grid-energy balance.



The horizon-sensitivity plot shows that observed-power smoothing is mostly controlled by the smoothing window rather than forecast horizon, because it does not use forecasts. Longer smoothing windows give stronger ramp reduction, with gradually higher storage power and energy requirements.

The forecast-informed target gives stronger ramp reduction than observed-power smoothing, but it also increases the implied storage burden. This happens because the storage system must compensate not only for smoothing, but also for forecast mismatch. The increase is much smaller than in the uncertainty-aware case, but it is clearly larger than smoothing the observed WEC signal directly.

The uncertainty-aware target becomes more conservative as the forecast horizon increases. The lower prediction bound moves farther below the realized WEC power, so the grid/raw energy ratio decreases and the required energy rises. This highlights the cost of using uncertainty to reduce under-delivery risk: the grid target becomes smoother and safer, but more surplus or curtailment-like energy appears.

Overall, forecasting can support smoother grid-export targets, but forecast errors and bias transfer part of the forecasting problem into the storage system. Prediction intervals add useful risk awareness, but a lower-bound target should be interpreted as a conservative commitment rule rather than a direct BESS-sizing recommendation.

7. Representative high-ramp case study

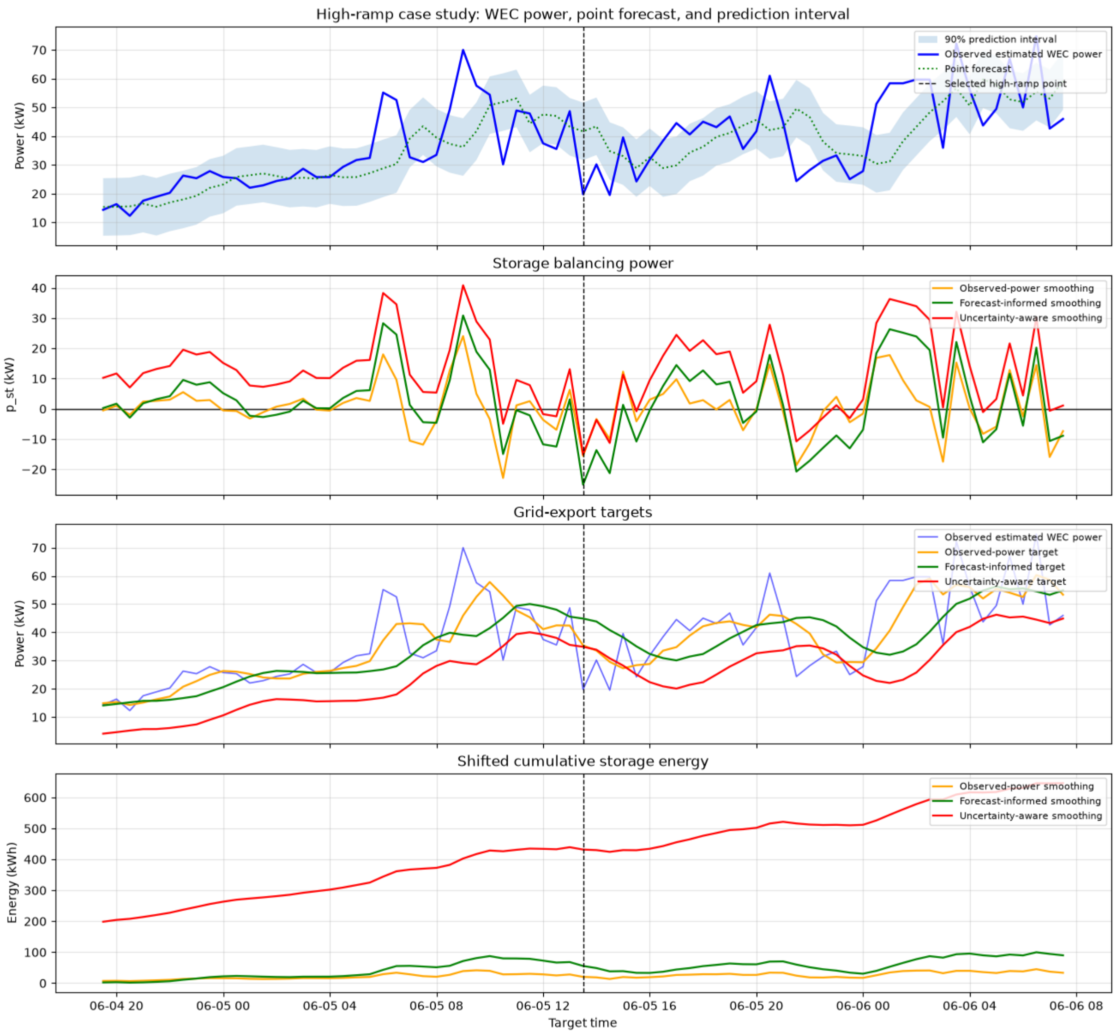
The previous sections compared scenario metrics across smoothing windows and forecast horizons. This section shows one representative high-ramp period in time-series form.

The example uses the 30 min forecast horizon and a 2 h smoothing window. The selected time window is centered around the largest absolute WEC power ramp in the aligned 30 min test rows.

The plot compares:

- observed estimated WEC power
- point forecast and 90% prediction interval
- observed-power, forecast-informed, and uncertainty-aware grid-export targets
- corresponding storage balancing power
- cumulative storage energy trajectory

This visual example helps show how the grid-export target definition changes the storage response during a rapidly varying period.



Case-study center time: 2025-06-05 13:30:00
Forecast horizon: 30 min
Fold: 2
Segment: h1_fold2_seg4
Smoothing window: 2 h
Context before/after: 18 hours
Selected ramp magnitude: 28.89 kW per 30 min

The high-ramp case study shows how the same observed WEC power period leads to different storage responses depending on the grid-export target definition.

The observed-power target follows the WEC signal most closely, so its storage balancing power remains smaller and alternates around zero. The forecast-informed target is smoother, but it does not always match the timing and level of the observed WEC power. This creates larger charge and discharge excursions, showing how forecast mismatch can transfer into storage operation.

The uncertainty-aware target remains lower because it is based on the lower prediction bound. This reduces the committed grid-export level, but it also creates a persistent positive storage balance during much of the example window. The steadily increasing cumulative energy trajectory should therefore be interpreted as surplus or curtailment-like energy under a conservative target, not as a recommended battery size.

Overall, the example illustrates the main mechanism behind the summary metrics: smoothing reduces grid-export variability, but the choice of target determines whether the storage mainly shifts energy through time, compensates for forecast mismatch, or absorbs surplus relative to a conservative uncertainty-aware commitment.

8. Summary and limitations

This notebook evaluated simple storage-aware smoothing metrics for estimated WEC power on the 250 kW interpretation scale. The goal was not to optimize BESS dispatch, but to quantify how different grid-export target definitions affect ramp reduction, storage power, usable energy, throughput, and energy balance.

The results show the expected smoothing trade-off. Longer smoothing windows reduce p95 grid-export ramps, but require larger storage power and energy. Observed-power smoothing provides the cleanest smoothing baseline because the grid/raw energy ratio stays close to 1, meaning the target mainly shifts energy through time rather than changing the total committed energy.

Forecast-informed smoothing achieves stronger ramp reduction than observed-power smoothing, but it also increases storage needs because the storage system must compensate for forecast mismatch. The uncertainty-aware lower-bound target is more conservative and reduces the committed grid-export level, but it creates more surplus or curtailment-like energy. This makes it useful as a risk-aware illustration, not a direct BESS-sizing recommendation.

A balanced illustrative case from the results is the 30 min forecast horizon with a 2 h observed-power smoothing target. For the generic 250 kW WEC proxy, this observed-power smoothing case reduces the p95 ramp by about 70%, while implying roughly 31 kW of storage power rating and 52 kWh of usable storage energy. This is treated as an example smoothing-implied buffer requirement for this specific metric scenario, not an optimized BESS design point.

Illustrative linear scaling of the 2 h observed-power smoothing case

Number of 250 kW WECs	Total WEC rating (MW)	Storage power rating (MW)	Usable storage energy (MWh)
1	0.25	0.03	0.05
5	1.25	0.15	0.26
10	2.5	0.31	0.52
20	5	0.62	1.04

The array-scaling table linearly multiplies the single-device smoothing requirement to give an approximate order-of-magnitude interpretation for multiple 250 kW WEC units. This is only a simple scaling exercise. A real WEC array would not necessarily scale linearly because devices may experience different wave phases, spatial smoothing, electrical constraints, hydrodynamic interactions, wake effects, control strategies, and curtailment rules. The table should therefore be read as a transparent first-order estimate, not as a WEC-farm storage design.

Several limitations remain. The WEC power signal is estimated from a simplified power-matrix approach, not measured from an operating device. The 250 kW scale is a generic interpretation proxy. The smoothing rules are simple metric scenarios, not optimized storage dispatch or grid-code-compliant control. Storage is treated as an ideal balancing buffer, without converter limits, round-trip efficiency, degradation, thermal constraints, market participation, or detailed BESS/HESS control.

The prediction intervals are empirical forecast-uncertainty bands, not physical bounds on WEC power production. The lower-bound uncertainty-aware target is useful for illustrating conservative grid commitment, but it should not be interpreted as a validated operational policy. Overall, the results should be read as storage-relevant grid-integration indicators that connect WEC power variability, forecasting, uncertainty, and smoothing requirements.